

NOVAPAY

Fraud Detection Initiative

Business Impact & Executive Summary

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Prepared for: NovaPay Leadership & Business Stakeholders

Companion to the full Technical Report

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1. Executive Summary

NovaPay set out to replace its static, rules-based fraud detection system with a machine-learning system that catches more fraud, wrongly blocks fewer legitimate customers, and gives compliance teams a clear, auditable explanation for every decision. This report summarizes what was delivered, what it means for the business, and what NovaPay's leadership should know before relying on it in production.

The short version: the new system catches meaningfully more fraud than the old rules-based approach, does so while keeping false alarms low, explains every flagged transaction in plain language for compliance review, and includes an automatic safeguard that refuses to promote a new model version unless it actually clears the fraud-catching bar the business set — rather than trusting a purely statistical score.

+16.5% more fraud caught than the old rules-based system	92% of flagged transactions are genuinely fraudulent	Automatic quality gate before any model reaches production
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What this means for the business

Every number in this report reflects a system that was actually built and tested, not a proposal. The fraud-catching improvement, the explainability delivered to compliance, and the automatic safety checks described here were all verified working before this report was written.

2. Why This Matters to NovaPay

NovaPay is a digital-first, cross-border money transfer company operating across the United Kingdom, Canada, and the United States, processing millions of transactions every month. The business is built on three pillars: affordability, fast transfers, and a smooth digital experience — and it serves people for whom a delayed or wrongly-blocked transaction is not a minor inconvenience: expatriates supporting family back home, businesses running international payroll, and freelancers waiting on payment from clients abroad.

Fraud detection sits at the intersection of all three pillars. Missed fraud is a direct financial loss and a trust failure. A false alarm that blocks a legitimate customer's transfer undermines the exact speed and reliability NovaPay is built to deliver. And because fraud losses, chargebacks, and compliance penalties are costs that ultimately have to be absorbed somewhere, weak fraud detection puts quiet, ongoing pressure on the affordability NovaPay promises its customers. Accurate, fast, explainable fraud detection is not a back-office technical concern here — it is a direct input into whether NovaPay can keep its core promises to the people who depend on it.

3. The Business Challenge

The previous system relied on static, manually-configured rules combined with human review — an approach that struggles on three fronts at once:

3.1 Operational Challenges

- Real-time transaction processing leaves little room for manual review at NovaPay's transaction volume.
- Static rules do not adapt as fraud tactics evolve — identity theft, account takeover, transaction laundering, and unauthorized payments are constantly-shifting threats, and a fixed rule written last year does not automatically account for a new tactic that emerged last month.

3.2 Financial Challenges

- Refunds, chargebacks, and compliance penalties from missed fraud erode profitability directly.
- False positives — legitimate transactions wrongly blocked — frustrate customers and can drive them to a competitor, which is a slower-moving but equally real cost.
- Missed fraud does not just cost the value of the fraudulent transaction itself; it also increases the operational cost of the investigations and disputes that follow.

3.3 Regulatory Challenges

- Financial regulators require fraud detection systems to be transparent and auditable, not opaque black boxes.
- Anti-Money-Laundering (AML) and Know-Your-Customer (KYC) frameworks specifically demand that decisions be explainable — a compliance officer needs to be able to answer “why was this transaction flagged?” in plain language, not just point at a score.
- Inadequate fraud controls carry real regulatory penalty risk and invite increased scrutiny of the business more broadly.

One further finding worth flagging directly: initial planning assumed fraud represented under 1% of NovaPay's transactions. Analysis of the actual transaction data found the real rate closer to 9% — still a minority of transactions, but a materially different picture than originally assumed. This mattered in practice: the techniques appropriate for an extremely rare (sub-1%) problem are different from, and can actually work worse than, the techniques appropriate at NovaPay's real prevalence. Correcting this assumption early avoided building a more complex, harder-to-maintain solution engineered for a rarer problem than NovaPay actually has.

4. What We Built — In Plain Terms

Rather than a single model file handed off with a README, the delivered system is a complete, working pipeline covering everything from raw transaction data to a live decision service a compliance officer or a fraud analyst can actually use day to day:

- **A cleaned, well-understood dataset.** Every data-quality issue found — typos in category fields, inconsistent formatting, duplicate records — was investigated and resolved with a documented reason, not silently patched over.
- **A fraud-scoring engine.** Several candidate approaches were trained and compared honestly against the old rules-based system on the same data, so the improvement reported in this document is measured, not assumed.
- **Plain-language explanations for every flagged transaction.** Instead of a bare risk score, each flag comes with the specific, real-world reasons behind it — e.g., “new account, high transaction velocity in the past 24 hours, unusually risky sending corridor” — in the transaction's actual values, not statistical jargon.
- **A live scoring service.** Transactions can be scored in real time as they happen, and in batches for retrospective review, without manual intervention.
- **An analyst console.** A visual, no-code interface fraud-ops and compliance staff can use directly — score a transaction, review a batch, see the model's current performance, and check for early warning signs the system's environment has changed — without needing engineering support for routine use.
- **Automated early-warning monitoring.** The system watches for signs that incoming transaction patterns, or its own decisions, have started drifting away from what it was trained on — an early signal that retraining may be needed before performance quietly degrades.
- **A one-command way to run the whole thing.** The entire system — scoring service and analyst console together — can be started with a single command, lowering the barrier for anyone on the team to run and evaluate it.

5. Business Results

5.1 More Fraud Caught, Fewer Legitimate Customers Blocked

Measured on transaction data the system had never seen during development — the fairest test of how it would behave in production:

What we measured	Old rules-based system	New ML system	What this means
Fraud caught (out of every 100 fraud cases)	81	94	13 more real fraud cases caught per 100, without changing how many transactions get flagged for review
Accuracy of each flag (real fraud vs. false alarm)	67 in 100 flags were real fraud	92 in 100 flags were real fraud	Far fewer legitimate customers wrongly blocked, for a similar volume of review work

The 16.5% improvement in fraud caught exceeds the project's own success threshold (a minimum 15% improvement). Just as importantly, this was not achieved by simply flagging more transactions and accepting more false alarms — the accuracy of each individual flag improved substantially at the same time, which is the combination that actually matters for both loss prevention and customer trust.

5.2 An Automatic Safeguard Against Deploying a Weak Model

A machine learning system like this one is not a single fixed program — it is periodically retrained as new transaction data and new fraud patterns become available, and each retraining produces a new candidate to potentially replace the current one. This creates a real business risk if left unmanaged: a new candidate can look statistically impressive on paper while actually performing worse on the metric the business actually cares about.

To guard against this, the system now includes an automatic checkpoint: before any candidate can be selected, it must first prove it beats the old rules-based system's fraud-catching rate by at least the required 15% margin. Only candidates that clear this bar are even considered for promotion; among those, the strongest overall performer is chosen automatically. If, on some future retraining run, none of the candidates clear the bar, the system does not silently pick the least-bad option and move on — it flags this loudly for a human to review before anything changes in production.

Why this matters in practice

This checkpoint was added after discovering that three of the four candidate approaches evaluated during development actually fell just short of the required improvement (14.1% versus the required 15%) — a gap that a purely statistical comparison would not have caught on its own. The system now checks for this automatically on every future retraining, not just this one.

6. Regulatory & Compliance Value

AML and KYC frameworks require that fraud decisions be explainable, not just accurate. This system was built with that requirement as a first-class feature rather than an afterthought: every flagged transaction is accompanied by the specific factors that drove the decision, expressed in the transaction's real, human-readable values — not an abstract statistical score a compliance officer would need a data scientist to interpret.

- A compliance officer can answer “why was this transaction flagged?” directly from the system's own output, with real values like account age, transaction velocity, or corridor risk — not a black-box probability.
- Every model version's performance, including which candidates were considered and rejected and why, is recorded and auditable — supporting the kind of documentation regulators expect when reviewing fraud-control adequacy.
- A documented process exists for retraining, promoting, and rolling back model versions — including a sign-off record for each promotion decision — so “which model was live, when, and why” is always answerable.

7. Customer Experience & Competitive Advantage

The customer-facing impact of the false-positive improvement (Section 5.1) is direct: fewer expatriates, business owners, and freelancers experience the frustration of a legitimate transfer being wrongly blocked or delayed for manual review. Since NovaPay's customer satisfaction strategy depends heavily on secure, frictionless transaction processing, this is not a peripheral technical improvement — it supports the core experience NovaPay is trying to deliver.

More broadly, machine-learning-driven fraud detection is increasingly the standard among competing digital-first payment platforms. Modernizing NovaPay's approach is as much about keeping pace with the sector's baseline expectations as it is about the specific numbers reported here.

8. Operational Efficiency for the Fraud Team

Real-time automated scoring means the vast majority of transactions are evaluated without any manual review at all, freeing the fraud operations team to focus their attention on the smaller set of genuinely high-risk cases the system flags — rather than spreading review effort thinly across a much larger and less-targeted volume. The analyst console gives the team direct, self-serve access to score individual transactions, review batches, and monitor the system's health, without needing to file a request with engineering for routine questions.

9. Risk Management, Limitations & What's Left To Do

An honest account of what this system does not yet cover is as important as what it does — leadership should sign off on this initiative with these points in view, not discover them later:

- **The system needs periodic retraining.** Fraud tactics evolve, and a model trained on last year's patterns will gradually lose effectiveness. A documented retraining process exists (Section 6), but it requires an ongoing operational commitment, not a one-time setup.
- **Early-warning monitoring has been tested under simulated conditions, not yet live production traffic.** The monitoring logic has been verified to correctly detect both an unchanged environment (no false alarms) and a substantially changed one (correctly triggers), but it has not yet observed real, live production drift, since the system has not yet been deployed to production.
- **Containerized deployment (the standard way of running this kind of service in production) has been fully built and prepared but not yet tested end-to-end in a live infrastructure environment.** This final verification step requires production-like infrastructure that was not available during development, and should be completed before go-live.
- **The current decision threshold is a single, global setting.** Early analysis suggests fraud risk varies meaningfully by transaction corridor and channel; a future refinement could tune this more precisely rather than applying one setting uniformly, which may further improve results.
- **The system does not yet include user authentication or rate-limiting on its live service.** These are standard production-hardening steps that should be added as part of the go-live process, not gaps in the fraud-detection logic itself.

10. Recommended Next Steps

- **Run a shadow-mode pilot.** Deploy alongside the existing rules-based system, scoring live traffic without acting on it, to validate real-world performance before cutover.
- **Complete infrastructure deployment testing.** Finish the end-to-end verification of the containerized deployment in a production-like environment.
- **Establish the retraining cadence and ownership.** Assign clear ownership of the quarterly (at minimum) retraining and promotion-review process described in this report.
- **Add production-hardening steps.** Authentication, rate-limiting, and standard operational safeguards ahead of handling real customer traffic.
- **Review corridor- and channel-specific thresholds.** As a fast-follow improvement once live data is available, to further reduce false positives in lower-risk segments.

11. Conclusion

This initiative delivers what the business asked for: a fraud detection system that catches meaningfully more fraud than the legacy rules-based approach, does so while reducing false alarms rather than trading one for the other, gives compliance and fraud-ops teams the transparent, auditable explanations regulators and internal governance require, and includes a built-in safeguard against ever promoting a model that fails to meet the standard the business set for it. The remaining work — shadow-mode validation, final infrastructure testing, and standing up an ongoing retraining cadence — is clearly scoped and does not represent open questions about whether the approach works, but the operational steps required to run it responsibly in production.